

Presentation Script

Factor Models of Interest Rate Curves

Tightened speaker notes based on the report and slide deck

Slide 1 – Title

We present our study of factor models for the U.S. Treasury yield curve, focusing on the Nelson–Siegel framework and its dynamic extension. We evaluate cross-sectional fit, parameter stability, and forecasting performance relative to simple statistical benchmarks.

Slide 2 – Outline

We begin with the Nelson–Siegel model and its economic interpretation.

Slide 3 – NS Yield: Derivation

The Nelson–Siegel model starts from a parametric form for the instantaneous forward rate, written as a combination of level, slope, and curvature terms. Integrating that forward curve gives a zero-coupon yield that is linear in three components: a constant term, a decaying term, and a hump-shaped term.

Economically, β_0 captures the long-run level, β_1 controls the short end relative to the long end, β_2 captures medium-maturity curvature, and λ determines the timescale, or where the hump appears.

Slide 4 – NS Loadings and Example Curve

This slide shows the two Nelson–Siegel loading functions and an example yield curve. The first loading decays monotonically from one to zero, so it governs how the short end converges to the long-run level. The second loading is hump-shaped, peaking around $1/\lambda$, so it controls curvature at intermediate maturities. In the example, the negative slope factor produces an upward-sloping curve, while the positive curvature factor adds a hump in the middle of the maturity spectrum.

Slide 5 – Data

Slide 6 – GSW Daily Yield Panel

We construct a daily panel of U.S. Treasury zero-coupon yields from the GSW (Gurkaynak–Sack–Wright) data set. The sample runs from 2016 to early 2026 and contains 2,524 trading days across eight maturities, from 3 months to 10 years. We use zero-coupon yields implied by the Svensson fit, which is better aligned with discount-curve modeling and avoids coupon-related distortions.

This figure shows the daily yield panel over the full sample. The data cover several distinct regimes, including the 2016 to 2019 hiking cycle, the near-zero COVID period, the 2022 to 2023 inversion episode, and the later easing and steepening phase. This gives a useful test environment because the model has to remain stable across very different curve shapes and policy regimes.

Slide 7 – NS Fit on Selected Dates

This slide previews the observed curves and fitted Nelson–Siegel curves on several representative dates. In calm periods, which are the top two panels, all methods fit well, but in stressed periods like the bottom two

panels the choice of λ becomes much more important. These examples motivate the calibration results in the next section.

Slide 8 – Cross-Sectional Calibration

Slide 9 – Fit Quality: Three Lambda Methods

For each day, we fit the Nelson–Siegel curve by minimizing the sum of squared errors across the eight maturities. Conditional on λ , the problem is linear in the three beta coefficients, so those can be estimated by OLS.

We compare three approaches for choosing λ : daily grid search with conditional OLS, a fixed λ and full nonlinear optimization. To select a fixed λ median RMSE across all days, we looked at the median in-sample RMSE. Its minimum is broad and flat, so the fit quality is not very sensitive to the exact choice of $\lambda \approx 0.41$.

The full nonlinear optimization is chosen to be L-BFGS-B, which is efficient for smooth objectives. Grid search and nonlinear optimization deliver nearly identical fit quality, with median RMSE around 0.0016 percent. This means the grid is already fine enough and nonlinear optimization adds essentially no value. By contrast, the fixed- λ approach performs much worse, especially during stressed periods.

Slide 10 – Time Series of Lambda

This figure shows the time series of the estimated λ . In normal periods, λ is relatively stable, but it rises sharply during the 2022 to 2023 inversion episode. That means the effective hump in the Nelson–Siegel loadings shifts toward shorter maturities.

Also, zoomed in on the graph, in the first half of 2018, the nonlinear optimization shows instability in λ while the grid search remains more stable; in the second half, both methods produce elevated but stable values, coinciding with the September and December rate hikes.

Slide 11 – Daily RMSE: Three Methods

This slide shows the daily RMSE paths for the three methods. The grid-search and nonlinear-optimization curves almost completely overlap, confirming that they are effectively equivalent in practice. The fixed- λ method tracks them in calm markets, but its errors spike sharply during stress, particularly during inversion. So a fixed decay parameter is not reliable when curve fitting matters most.

Slide 12 – Representative Curve Fits

These selected curve fits illustrate the same point more clearly.

In the top two panels, which correspond to relatively stable market conditions, all three methods produce very similar fits. The curve shapes are smooth and well-behaved, and even a fixed λ provides sufficient flexibility to match the cross section closely. As a result, the RMSE differences across methods are very small in these periods.

In contrast, the bottom two panels correspond to stressed regimes, particularly during the 2022–2023 hiking cycle and inversion. Here, the shape of the yield curve changes significantly, with a much steeper front end and, in some cases, inversion at longer maturities.

In these cases, the fixed- λ specification cannot adjust the location of the curvature hump, so it fails to match both the short end and the long end simultaneously. This leads to visibly poorer fits, especially around

intermediate maturities where curvature is most pronounced. Its errors are about 20 to 30 times larger than those of the adaptive methods.

The adaptive methods are able to adjust λ upward, effectively shifting the hump. This allows them to capture the changing shape of the curve and maintain a tight fit even under stress.

Slide 13 – NS Factor Time Series

Here we plot the estimated Nelson–Siegel factors over time using the grid-search method. The broad trends are economically intuitive: the level factor rises during the hiking cycle, the slope factor turns positive during inversion, and the curvature factor becomes more volatile in stressed regimes. However, there are also extreme spikes, especially in 2022 to 2023, and these are clearly not economically plausible. Extreme spikes in 2022–23 are numerical artefacts from multiple near-equivalent OLS solutions — addressed by the Kalman filter in Part D.

Slide 14 – Factor Volatility and Correlation

The volatility and correlation diagnostics confirm that instability. Factor volatility rises sharply during the stressed period and then falls back afterward. We also see a moderate negative correlation between level and curvature, suggests that when the long-run level rises, estimated curvature tends to fall.

This also reflects the tradeoff that different parameter combinations can produce very similar yield curves. During stressed periods, this leads to multiple near-equivalent solutions, which explains the large parameter instability we observe. Therefore, daily cross-sectional fitting alone is not enough to stabilize the factors.

Slide 15 – Parameter Instability: Beta_0 > 20 Outliers

We looked more closely at the extreme outliers where the estimated level factor exceeds 20 percent, which is not economically reasonable. These occur because the objective function can admit two near-equivalent local minima on the same day. We tested two ad hoc fixes: dropping the very short maturities, and imposing parameter bounds in constrained nonlinear optimization. Both remove the outliers, but each introduces a tradeoff, either by discarding short-end information or by slightly worsening fit.

Slide 16 – Lambda and Beta_0: Before and After Remedies

This figure shows the time series of lambda and beta_0 before and after those remedies. The remedies smooth the most extreme spikes, but they do not solve the underlying identification problem. So while they are useful diagnostics, they are not a fully satisfactory solution. That motivates moving to a dynamic state-space approach.

Slide 17 – Dynamic NS via Kalman Filter

Slide 18 – Filtered vs OLS vs Nonlinear Factors

In the dynamic Nelson–Siegel model, the latent factors follow a VAR(1) process and are estimated jointly over time using the Kalman filter. The measurement equation links observed yields to the factors through the Nelson–Siegel loadings, and the state equation imposes time-series structure on the factors. This adds a dynamic prior that penalizes abrupt and implausible jumps while still allowing the model to adapt over time.

This comparison shows the filtered factors against the OLS and nonlinear daily estimates. The main difference is that the Kalman filter removes the abnormal spikes entirely. Because the filtered estimate must remain consistent with the VAR dynamics, it stays economically sensible.

Slide 19 – Beta_0 Around the Outlier Period

This slide zooms in on the problematic 2023 period. Under OLS and nonlinear estimation, the level factor jumps to extremely high values, but under the Kalman filter it remains in a narrow and economically reasonable range. Across the full sample, the Kalman estimate of beta_0 stays below about 6 percent, compared with values above 20 percent in the static methods. So the dynamic prior is very effective at stabilizing the factors.

Slide 20 – In-Sample RMSE: OLS vs Kalman

The in-sample fit statistics show that the Kalman filter also improves fit quality substantially. When all methods are evaluated using the same fixed loading matrix, the Kalman filter reduces median RMSE by about a factor of 10 relative to the two-step methods. Its maximum RMSE is also much lower. So the filter improves both economic plausibility and statistical fit.

Slide 21 – Out-of-Sample Forecast RMSE: DNS vs Random Walk

We then test the dynamic model in one-day-ahead out-of-sample forecasting. The comparison here is between the DNS–Kalman forecast and a random-walk benchmark. The result is that DNS performs worse: its median forecast RMSE is more than twice as large as the random walk. So although the dynamic model improves factor estimation, that does not translate into better short-horizon forecasts than random walk.

Slide 22 – Cumulative Forecast Accuracy: DNS vs Random Walk

This slide shows cumulative squared forecast error over time. The DNS forecast remains consistently above the random walk, with a cumulative SSE ratio of about 2.53. The main reason is that the estimated VAR is extremely persistent, with a dominant eigenvalue near 0.996. At a one-day horizon, that makes the model behave like a noisy random walk, but with added estimation error. That problem becomes even worse during strongly trending regimes such as 2022 to 2023.

Slide 23 – VAR(1) Impulse Response Functions

The IRFs reinforces that level shocks are nearly permanent, slope shocks decay over weeks, and curvature shocks mean-revert quickly, confirming the persistence in the yield curve factors. The near-unit-root behavior of the level factor explains why the DNS model behaves like a noisy random walk at short horizons, which is consistent with its poor out-of-sample forecasting performance.

Slide 24 – Forecasting Yield Changes and Trading

In Part E, we move further from forecasting yield levels to forecasting yield changes in the 2-year, 5-year, and 10-year maturities at 1-day, 1-week, and 1-month horizons. We compare three models: a random walk for changes, univariate AR(1) models, and DNS–Kalman forecasts produced from factor dynamics. Across all maturities and horizons, the random-walk and AR(1) benchmarks outperform DNS–Kalman in RMSE terms. So the dynamic factor model does not add predictive value for directional yield forecasting.

Slide 25 – Conclusion

We conclude with four main findings. First, grid search and nonlinear optimization are effectively equivalent, so a fine grid is sufficient for estimating λ . Second, a fixed λ works in calm periods but fails badly under stress, so λ must be allowed to vary over time. Third, the Kalman filter substantially improves factor stability and in-sample fit by imposing dynamic structure and eliminating pathological solutions. Fourth, despite these improvements, DNS–Kalman does not outperform simple benchmarks in short-horizon forecasting, so its main value lies in curve fitting, interpolation, pricing, and risk factor extraction rather than directional trading.

Slide 26 – Thank You

Thank you.

The dataset provides daily Svensson (1994) parameters ($\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2$) fitted to the outstanding Treasuries daily. This model has more parameters and greater flexibility than the NS model, so it can be used to construct a clean panel of zero-coupon yields at arbitrary maturities. As the Svensson fit matches the observed Treasury market data closely (error is below 1 basis point, as noted in the paper), the implied zero-coupon yield curve is a reliable synthetic discount curve constructed from those bond prices.

The random walk outperforms DNS because interest rates are highly persistent and close to a unit root, so tomorrow's yield is best predicted by today's yield. The DNS model adds weak mean reversion and estimation noise, which actually worsens short-horizon forecasts.